

SQL Interview Short Questions and Answers :

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Question 1] What is SQL?

Answer] SQL (Structured Query Language) is used to store, retrieve, manipulate, and manage data in relational databases. It allows analysts to query large datasets efficiently.

Question 2] What are the different types of SQL commands?

Answer]

- **DDL** – CREATE, ALTER, DROP
- **DML** – INSERT, UPDATE, DELETE
- **DQL** – SELECT
- **DCL** – GRANT, REVOKE
- **TCL** – COMMIT, ROLLBACK, SAVEPOINT

Question 3] Difference between WHERE and HAVING?

Answer]

WHERE

Filters rows

Used before GROUP BY

Cannot use aggregates

HAVING

Filters groups

Used after GROUP BY

Can use aggregates

Question 4] What is a Primary Key and a Foreign Key?

Answer]

- **Primary Key:** Uniquely identifies each record (no NULLs).
- **Foreign Key:** References the primary key of another table to maintain relationships.

Question 5] What is the difference between DELETE, TRUNCATE, and DROP?

Answer]

DELETE

Removes rows

Can rollback

Slower

TRUNCATE

Removes all rows

Cannot rollback

Faster

DROP

Deletes table

Permanent

Removes structure

Question 6] What are the different types of JOIN in SQL?

Answer]

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- SELF JOIN

Question 7] What is the use of GROUP BY and ORDER BY?

Answer]

- **GROUP BY:** Groups rows for aggregation
- **ORDER BY:** Sorts result set (ASC/DESC)

Question 8] What is a NULL value and how is it different from 0 or blank?

Answer]

- **NULL** means unknown/missing
- **0** is a numeric value
- **Blank** is an empty string

Question 9] What is a Subquery? How is it different from JOIN?

Answer]

- **Subquery:** Query inside another query
- **JOIN:** Combines tables directly

JOINS are faster and preferred for large datasets.

Question 10] What is a constraint? Name a few types of constraints?

Answer] Constrains are the rules enforced on data integrity

- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- NOT NULL
- CHECK
- DEFAULT

Question 11] Write a query to find the second highest salary from an employee table.

Answer]

```
SELECT MAX(salary)
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

Question 12] What is the difference between IN and EXISTS?

Answer]

IN	EXISTS
Compares values	Checks condition
Slower for large data	Faster
Uses list	Uses subquery

Question 13] What is a UNION and how is it different from UNION ALL?

Answer]

- **UNION** removes duplicates
 - **UNION ALL** keeps duplicates (faster)
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Question 14] What are indexes? How do they improve performance?

Answer] Indexes are data structures that speed up data retrieval in databases by creating sorted lookups, similar to a book's index, allowing the database to find specific records quickly without scanning every row.
It improves the performance by speeding up data retrieval by creating pointers to rows.

Question 15] Explain normalization. What are 1NF, 2NF, 3NF?

Answer] Normalisation is the process to reduce redundancy.

- **1NF**: Atomic values
 - **2NF**: No partial dependency
 - **3NF**: No transitive dependency
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Question 16] What is a view? How is it different from a table?

Answer] View is a **virtual table** based on a SQL query. Stores query logic, not actual data.

View is different from table as table is a physical database object that directly stores data in rows and columns. Tables are independent and occupy storage, whereas views are dependent on tables.

Question 17] What is the difference between CHAR and VARCHAR?

Answer]

CHAR	VARCHAR
Fixed length	Variable length
Faster	Space efficient

Question 18] What are aggregate Functions in SQL ? Give examples

Answer] Aggregate functions in SQL are the functions used for calculations on multiple rows.

Examples:

- SUM()
 - COUNT()
 - AVG()
 - MIN()
 - MAX()
-

Question 19] What is a self join?

Answer] Joining a table with itself using aliases, useful for hierarchical data.

Question 20] Write a query to get the top 3 departments with the highest number of employees.

Answer]

```
SELECT department, COUNT(*) AS emp_count
FROM employee
GROUP BY department
ORDER BY emp_count DESC
LIMIT 3;
```

Question 21] Write a query to count the number of employees in each department.

Answer]
SELECT department, COUNT(*)
FROM employee
GROUP BY department;

Question 22] Retrieve names of employees who joined in the last 6 months.

Answer]
SELECT name
FROM employee
WHERE joining_date >= CURRENT_DATE - INTERVAL '6 months';

Question 23] Write a query to display duplicate records from a table.

Answer]
SELECT column_name, COUNT(*)
FROM table_name
GROUP BY column_name
HAVING COUNT(*) > 1;

Question 24] Get the maximum salary for each job role.

Answer]
SELECT job_role, MAX(salary)
FROM employee
GROUP BY job_role;

Question 25] Write a query to transpose rows into columns (using CASE OR PIVOT)

Answer]
SELECT
SUM(CASE WHEN month='Jan' THEN sales END) AS Jan,
SUM(CASE WHEN month='Feb' THEN sales END) AS Feb
FROM sales_table;